Skewness

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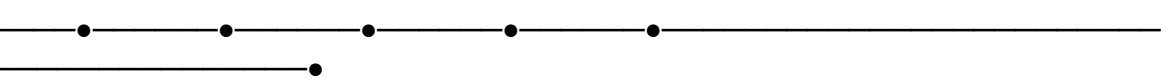
1. Daily Income of Food Delivery Agents

Given daily income data:

₹400, ₹420, ₹430, ₹450, ₹470, ₹500, ₹2,000

Number Line

₹400 ₹420 ₹430 ₹450 ₹470 ₹500 ₹2000



Most of the incomes are concentrated between **₹400 and ₹500**, while **₹2,000** is far away on the right side.

Observation:

The distribution has a **long tail towards the right**.

Answer: Right-Skewed Distribution

Reason: The unusually high income of ₹2,000 pulls the distribution towards the right side.

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2. Mean, Median and Skewness

Given income data:

₹350, ₹370, ₹400, ₹410, ₹415, ₹420, ₹2,000

Mean

Mean = Sum of all values ÷ Number of values

Mean = (350 + 370 + 400 + 410 + 415 + 420 + 2,000) ÷ 7

Mean = **₹3,965 ÷ 7**

Mean = ₹566.43

Median

The data is already arranged in ascending order:

₹350, ₹370, ₹400, ₹410, ₹415, ₹420, ₹2,000

There are 7 values, so the middle value is the **4th value**.

Median = ₹410

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Since:

Mean (₹566.43) > Median (₹410)

the distribution is **right-skewed**.

The high income of **₹2,000** is an extreme value that pulls the mean upward. Therefore, the mean is much higher than the median, indicating a **positive/right skewed distribution**.

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3. Monthly Pocket Money Dataset

Suppose the monthly pocket money received by 8 friends is:

₹1,500, ₹1,800, ₹2,000, ₹2,000, ₹2,200, ₹2,500, ₹2,700, ₹5,000

Mean

Mean = (1,500 + 1,800 + 2,000 + 2,000 + 2,200 + 2,500 + 2,700 + 5,000) ÷ 8

Mean = **₹19,700 ÷ 8**

Mean = ₹2,462.50

Median

Since there are 8 values, the median is the average of the 4th and 5th values.

Median = (₹2,000 + ₹2,200) ÷ 2

Median = ₹2,100

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Mean = **₹2,462.50**

Median = **₹2,100**

Since **Mean > Median**, the data is:

Right-Skewed

The ₹5,000 pocket money is much higher than the other values and pulls the mean towards the right.

What does this suggest?

Most friends receive pocket money between approximately **₹1,500 and ₹2,700**.

One friend receiving ₹5,000 makes the distribution **right-skewed**, showing that pocket money is not equally distributed among the group.

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4. Real-Life Example of a Skewed Dataset

Example: Instagram Followers

Consider the number of followers of different Instagram users.

For example:

100, 250, 400, 600, 1,000, 2,000, 50,000

Most users may have a relatively small number of followers, while a few users can have thousands or millions of followers.

Expected Distribution:

Right-Skewed Distribution

Reason: A small number of highly popular accounts have extremely large numbers of followers. These high values create a long tail on the right side of the distribution.

Final Summary

Dataset	Distribution	Main Reason
Food delivery income	Right-Skewed	₹2,000 is an unusually high value
Food delivery income – Q2	Right-Skewed	Mean is greater than median
Pocket money	Right-Skewed	₹5,000 is much higher than most values
Instagram followers	Right-Skewed	A few accounts have extremely high followers

Conclusion

Skewness describes the shape and direction of a distribution.

- **Right-skewed:** Long tail towards the right; generally Mean > Median.
- **Left-skewed:** Long tail towards the left; generally Mean < Median.
- **Symmetrical:** Data is balanced around the center; Mean ≈ Median.